GenAI assignment1

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Task 1: sigma algebra

1a

We are given the sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$ and the collection $\mathcal{E} = \{\{4\}, \{1, 2\}\}$. We want to find $\sigma(\mathcal{E})$, the smallest σ-algebra containing $\mathcal{E}$. To do this, we apply the three σ-algebra rules, starting from $\mathcal{E}$ and adding sets until no new sets need to be added.

A σ-algebra must always contain the sample space and the empty set, so we immediately include $\emptyset$ and Ω. Next, since a σ-algebra must be closed under complements, we add the complement of each set in $\mathcal{E}$: the complement of $\{4\}$ is $\{1, 2, 3, 5, 6\}$, and the complement of $\{1, 2\}$ is $\{3, 4, 5, 6\}$. We then close under unions: $\{4\} \cup \{1, 2\} = \{1, 2, 4\}$, which is a new set, so we must also include its complement $\{3, 5, 6\}$. Checking all remaining unions shows they produce sets already in our collection, so we are done.

The σ-algebra generated by $\mathcal{E}$ is:

$$\sigma(\mathcal{E}) = \{\emptyset,\ \{4\},\ \{1, 2\},\ \{1, 2, 4\},\ \{3, 5, 6\},\ \{3, 4, 5, 6\},\ \{1, 2, 3, 5, 6\},\ \Omega\}$$

Since any σ-algebra $\mathcal{F}$ containing $\mathcal{E}$ must contain $\{4\}$ and $\{1, 2\}$ by assumption, and is then forced to contain all remaining sets by closure under complements and unions, it follows that $\sigma(\mathcal{E}) \subseteq \mathcal{F}$ for any such $\mathcal{F}$. Hence $\sigma(\mathcal{E})$ is the smallest σ-algebra containing $\mathcal{E}$.

1b

Since Σ_1 and Σ_2 are both σ-algebras, they both contain $\emptyset$ and Ω, so $\emptyset, \Omega \in \Sigma$.

Let $A \in \Sigma$, meaning $A \in \Sigma_1$ and $A \in \Sigma_2$, since both are σ-algebras by the property of closed under compliments both also contain the A^c, and therefore so does Σ.

Let $A, B \in \Sigma$, meaning $A, B \in \Sigma_1$ and $A, B \in \Sigma_2$. Since both are σ-algebras, they are closed under unions, so $A \cup B \in \Sigma_1$ and $A \cup B \in \Sigma_2$, and therefore $A \cup B \in \Sigma$.

This shows that $\Sigma$is a σ-algebra over Ω. as Σ has all 3 properties of σ-algebra

1c

No Σ is not σ-algebra. Say we have a set $A \in \Sigma_1$, $B \in \Sigma_2$ and therefore $A, B \in \Sigma$, Σ should have $A \cup B$ since it should be closed under unions to be a σ-algebra, but this will not be the case since i general neither Σ_1, Σ_2 contain $A \cup B$.

Task 2: Expected values and Monte Carlo simulation

2a

1.

$$\sum_{k=0}^{6} p(X=k) = M\left(\sum_{k=0}^{4} \frac{4^k e^{-4}}{k!} + 2\right) = 1$$

$$M\left(\sum_{k=0}^{4} \frac{4^k e^{-4}}{k!} + 2\right) = 1.$$

$$M((1+4+8+10.667+10.667)e^-4)+2) = 1$$

$$M = 0.3804$$

2.

$$E[X] = \sum_{k=0}^{6} k\, p(X=k) = \sum_{k=0}^{4} k\, M \frac{4^k e^{-4}}{k!} + 5M + 6M$$

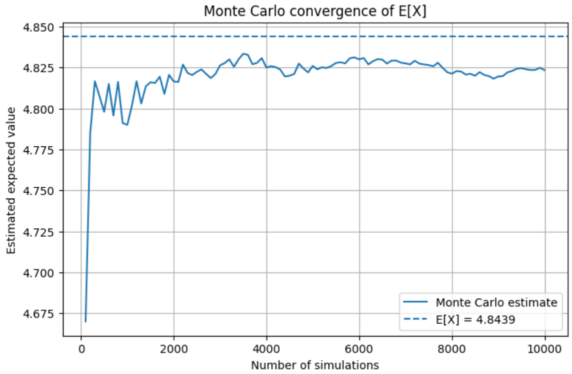
$$= M\left(\sum_{k=0}^{4} k \frac{4^k e^{-4}}{k!} + 11\right)$$

$$\sum_{k=0}^{4} k \frac{4^k}{k!} = 0 + 4 + 16 + 32 + \frac{1024}{24} = 94.6667$$

$$\sum_{k=0}^{4} k \frac{4^k e^{-4}}{k!} = 94.6667 \cdot e^{-4} \approx 1.734$$

$$E[X] = M(1.734 + 11) \approx 0.3804 \cdot 12.734 \approx 4.84$$

3.

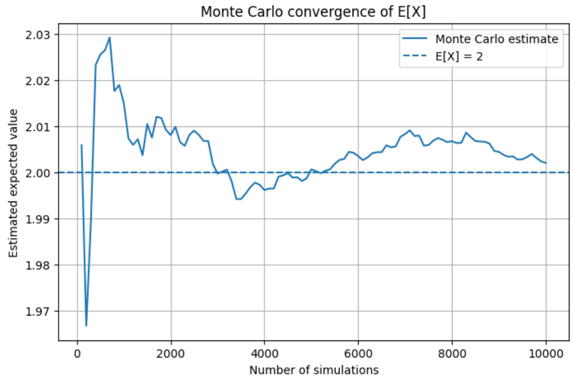


2b

Analytically:

$$E[X] = \int_0^3 \frac{2}{9}x^2 = 2$$

With the Monte Carlo:



2c

To find K, we choose a joint density for (X, Y). Since both integrals run from 0 to π, we choose a uniform distribution over $[0, \pi] \times [0, \pi]$:

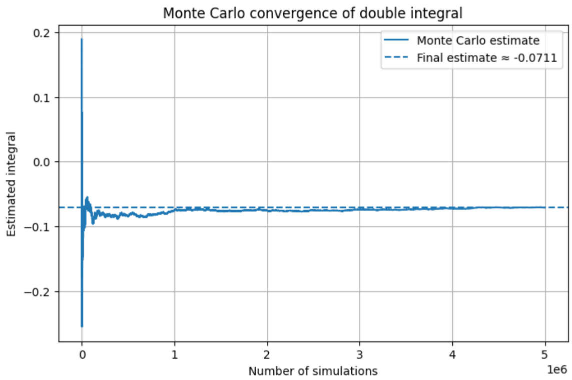
$$p(x, y) = \frac{1}{\pi^2}, \quad (x, y) \in [0, \pi] \times [0, \pi]$$

Multiplying and dividing the integral by π^2 gives:

$$\int_0^\pi \int_0^\pi \sin(2(\sqrt{x^2+y^2}+x+y))\,dx\,dy = \pi^2 \int_0^\pi \int_0^\pi \sin(2(\sqrt{x^2+y^2}+x+y)) \cdot \frac{1}{\pi^2}\,dx\,dy = \pi^2 \cdot E\Big[\sin(2(\sqrt{X^2+Y^2}+X+Y$$

Thus $K = \pi^2$.

With the Monte Carlo, we found that the final estimate for the Expected value was -0.071:



Task 3: Probability and Simulations

The word GLOOM contains 5 letters, where the letter 'O' appears twice. The total number of distinct permutations is given by:

$$\frac{n!}{k!} = \frac{5!}{2!} = \frac{120}{2} = 60$$

Since the two O's are indistinguishable, only 1 permutation out of 60 spells exactly GLOOM. Therefore the exact probability is:

$$P(\text{GLOOM}) = \frac{1}{60} = \frac{2}{120} \approx 0.0167$$

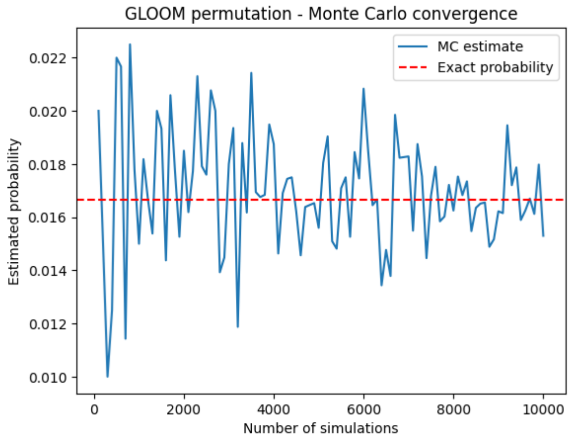


Figure 1: Enter Caption

Task 4: Entropy of a univariate Gaussian variable

Task 4

$$H[X] = -\int_{x \in \mathcal{X}} p_x(x) \ln(p_x(x))\,dx$$

$$= -\int_{-\infty}^{\infty} p(x) \ln\left(\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)\right) dx$$

$$= -\int_{-\infty}^{\infty} p(x) \left(\ln\left(\frac{1}{\sqrt{2\pi}\sigma}\right) + -\frac{1}{2\sigma^2}(x-\mu)^2\right) dx$$

$$= \int_{-\infty}^{\infty} p(x) \left(\ln\left(\sqrt{2\pi}\sigma\right) + \frac{1}{2\sigma^2}(x-\mu)^2\right) dx$$

$$= \int_{-\infty}^{\infty} p(x) \ln(\sqrt{2\pi}\sigma)\,dx + \int_{-\infty}^{\infty} p(x) \cdot \frac{1}{2\sigma^2}(x-\mu)^2\,dx$$

$$= \ln(\sqrt{2\pi}\sigma) + \frac{1}{2\sigma^2} \cdot \sigma^2 =$$

$$= \ln(\sqrt{2\pi}\sigma) + \frac{1}{2}$$

Task 5: Conditional Indipendence

Task 5a

We prove that the following conditions are equivalent:

1. $p(X, Y \mid Z) = p(X \mid Z)\, p(Y \mid Z)$
2. $p(X \mid Y, Z) = p(X \mid Z)$

(1 ⇒ 2) Assume condition 1 holds. By definition of conditional probability:

$$p(X \mid Y, Z) = \frac{p(X, Y \mid Z)}{p(Y \mid Z)}$$

Substituting condition 1 in the numerator:

$$p(X \mid Y, Z) = \frac{p(X \mid Z)\, p(Y \mid Z)}{p(Y \mid Z)} = p(X \mid Z)$$

(2 ⇒ 1) Assume condition 2 holds. By the chain rule:

$$p(X, Y \mid Z) = p(X \mid Y, Z)\, p(Y \mid Z)$$

Substituting condition 2:

$$p(X, Y \mid Z) = p(X \mid Z)\, p(Y \mid Z)$$

Task 5b

We want to prove that $X \perp\!\!\!\perp Y, W \mid Z$, i.e.:

$$p(X, Y, W \mid Z) = p(X \mid Z)\, p(Y, W \mid Z)$$

Applying the chain rule to the left-hand side:

$$p(X, Y, W \mid Z) = p(X \mid Y, W, Z)\, p(Y \mid W, Z)\, p(W \mid Z)$$

By assumption 1 ($X \perp\!\!\!\perp W \mid Z, Y$) and Task 5a:

$$p(X \mid Y, W, Z) = p(X \mid Y, Z)$$

By assumption 2 ($X \perp\!\!\!\perp Y \mid Z$) and Task 5a:

$$p(X \mid Y, Z) = p(X \mid Z)$$

Substituting back:

$$p(X, Y, W \mid Z) = p(X \mid Z)\, p(Y \mid W, Z)\, p(W \mid Z)$$

Finally, we recompose the last two terms using the chain rule:

$$p(Y \mid W, Z)\, p(W \mid Z) = p(Y, W \mid Z)$$

Therefore:

$$p(X, Y, W \mid Z) = p(X \mid Z)\, p(Y, W \mid Z)$$

Task 5c

Yes, the converse of Task 5b holds. If $X \perp\!\!\!\perp (Y, W) \mid Z$, then both $X \perp\!\!\!\perp Y \mid Z$ and $X \perp\!\!\!\perp W \mid Z, Y$ are satisfied. These are standard properties of conditional independence known as **Decomposition** and **Weak Union**.

1. Proof of $X \perp\!\!\!\perp Y \mid Z$ (Decomposition) We aim to show that $p(X, Y \mid Z) = p(X \mid Z)p(Y \mid Z)$. Starting from the definition of marginalization:

$$p(X, Y \mid Z) = \sum_w p(X, Y, W \mid Z) \tag{1}$$

By the hypothesis $X \perp\!\!\!\perp (Y, W) \mid Z$, we can factorize the joint term:

$$p(X, Y \mid Z) = \sum_w p(X \mid Z)p(Y, W \mid Z) \tag{2}$$

Since $p(X \mid Z)$ does not depend on w, we can move it outside the summation:

$$p(X, Y \mid Z) = p(X \mid Z) \sum_w p(Y, W \mid Z) \tag{3}$$

Applying the sum rule $\sum_w p(Y, W \mid Z) = p(Y \mid Z)$, we obtain:

$$p(X, Y \mid Z) = p(X \mid Z)p(Y \mid Z) \implies X \perp\!\!\!\perp Y \mid Z \tag{4}$$

2. Proof of $X \perp\!\!\!\perp W \mid Z, Y$ (Weak Union) We aim to show that $p(X \mid W, Y, Z) = p(X \mid Y, Z)$. From the hypothesis, we know:

$$p(X \mid W, Y, Z) = p(X \mid Z) \tag{5}$$

From the Decomposition property proved in the previous step, we also have $p(X \mid Y, Z) = p(X \mid Z)$. By substitution:

$$p(X \mid W, Y, Z) = p(X \mid Y, Z) \implies X \perp\!\!\!\perp W \mid Z, Y \tag{6}$$

Conclusion: Since both conditions are derived directly from the hypothesis via marginalization and the definition of conditional probability, the claim is proven.

Task 6: Urn problem

The probability that you draw exactly 2 different colors is 0.66077

Task 7: Marginal densities

7a

All possible outcomes

Outcome	X	Y
HHH	3	0
HHT	2	1
HTH	2	1
HTT	1	0
THH	2	1
THT	1	0
TTH	1	0
TTT	0	0

From that we determine out of the probabilities for all possible x and y.

$$p(0,0) = 1/8$$

$$p(1,0) = 3/8$$

$p(2,1) = 3/8$ $p(3,0) = 1/8$ The rest probabilities of the combinations of x and y are 0.

The marginals: $p(Y = 0) = 5/8$ $p(Y = 1) = 3/8$

7b

The sentence contains 7 words. Each word is picked with probability $\frac{1}{7}$. The words with their (X, Y) values are:

Word	X (length)	Y (# E's)
A	1	0
FAIR	4	0
COIN	4	0
IS	2	0
TOSSED	6	1
THREE	5	2
TIMES	5	1

Joint PMF $p(x, y)$:

	$Y=0$	$Y=1$	$Y=2$
$X=1$	$\frac{1}{7}$	0	0
$X=2$	$\frac{1}{7}$	0	0
$X=4$	$\frac{2}{7}$	0	0
$X=5$	0	$\frac{1}{7}$	$\frac{1}{7}$
$X=6$	0	$\frac{1}{7}$	0

Marginal PMFs:

$$p_X(1) = \frac{1}{7}, \quad p_X(2) = \frac{1}{7}, \quad p_X(4) = \frac{2}{7}, \quad p_X(5) = \frac{2}{7}, \quad p_X(6) = \frac{1}{7}$$

$$p_Y(0) = \frac{4}{7}, \quad p_Y(1) = \frac{2}{7}, \quad p_Y(2) = \frac{1}{7}$$

Independence check:

X and Y are independent if and only if $p(x,y) = p_X(x) \cdot p_Y(y)$ for all x, y. Taking $x = 5, y = 2$ as a counterexample:

$$p(5,2) = \frac{1}{7}$$
$$p_X(5) \cdot p_Y(2) = \frac{2}{7} \cdot \frac{1}{7} = \frac{2}{49}$$
$$\frac{1}{7} \neq \frac{2}{49}$$

Therefore X and Y are **not independent**.

Task 8: Sum of two random variables

Let X_1 and X_2 be two independent discrete random variables representing the outcome of two fair 6-sided dice. Each has a probability mass function $P(X_i = x) = \frac{1}{6}$ for $x \in \{1,2,3,4,5,6\}$. Let $Y = X_1 + X_2$.

The range of Y is $\{2, 3, \ldots, 12\}$. The total number of possible outcomes in the sample space is $6 \times 6 = 36$.

The number of ways to obtain a sum k corresponds to the number of integer pairs (x_1, x_2) such that $x_1 + x_2 = k$ with $1 \leq x_1, x_2 \leq 6$. This can be expressed using the following general formula:

$$P(Y = k) = \frac{6 - |k - 7|}{36} \quad \text{for } k \in \{2, 3, \ldots, 12\} \tag{7}$$

Task 9: Transformation of Random Variables

Task 9

$Q = A/B \in [0, 1]$ means $A \leq B$, so:

$$P(Q \in [0, 1]) = P(A \leq B) = \sum_{b=1}^{\infty} \sum_{a=1}^{b} P(A = a) \cdot P(B = b)$$

where independence allows us to multiply the probabilities.

Inner sum:

$$\sum_{a=1}^{b} P(A = a) = \sum_{a=1}^{b} 2^{-a} = 1 - 2^{-b}$$

Outer sum:

$$P(A \leq B) = \sum_{b=1}^{\infty} 2^{-b}(1 - 2^{-b}) = \sum_{b=1}^{\infty} 2^{-b} - \sum_{b=1}^{\infty} 4^{-b}$$

$$= \frac{1/2}{1 - 1/2} - \frac{1/4}{1 - 1/4} = 1 - \frac{1}{3} = \frac{2}{3}$$

Contributions

We made our best efforts to evenly split the work among all three members. Problems were divided in such a way that people were matched with topics that they had more prior experience with. We all had an involvement in coding for the problems through splitting sub questions. All collaboration was facilitated through a GitHub repository.

Here is a per question breakdown of the division:

- **Francesco Colasurdo**
  - Question 4
  - Question 5
  - Question 6
- **Keanu Lim A Po**

  - Question 1
  - Question 3
  - Question 8
- **Michael Rayan**
  - Question 2
  - Question 7
  - Question 9

Code

```
import numpy as np
import matplotlib.pyplot as plt
import math

#2a
x_values = np.array([0, 1, 2, 3, 4, 5, 6])

# first part of the pmf
poisson_part = sum((4**k * math.exp(-4)) / math.factorial(k) for k in range
    (5))

M = 1 / (poisson_part + 2)

probs = np.array([
    M * (4**k * math.exp(-4)) / math.factorial(k) if k <= 4 else M
    for k in x_values
])

Expectation = np.sum(x_values * probs)
n_sims= 10000
np.random.seed(42)
samples = np.random.choice(x_values,n_sims,p =probs)

simulation_sizes = np.arange(100, n_sims + 1, 100)

estimated_E = [np.mean(samples[:n]) for n in simulation_sizes]
plt.figure(figsize=(8, 5))
plt.plot(simulation_sizes, estimated_E, label="Monte Carlo estimate")
plt.axhline(y= Expectation, linestyle="--", label=f"E[X] = {Expectation:.4f}
    ")
plt.xlabel("Number of simulations")
plt.ylabel("Estimated expected value")
plt.title("Monte Carlo convergence of E[X]")
plt.legend()
plt.grid(True)
plt.show()
```

```
##2b
N = 10000
U = np.random.uniform(0, 1, N)
X = 3 * np.sqrt(U)

# Monte Carlo estimate

simulation_sizes = np.arange(100, 10001, 100)   # 100, 200, ..., 10000
E_est= [np.mean(X[:n]) for n in simulation_sizes]



plt.figure(figsize=(8, 5))
plt.plot(simulation_sizes, E_est, label="Monte Carlo estimate")
plt.axhline(y= 2, linestyle="--", label=f"E[X] = 2")
plt.xlabel("Number of simulations")
plt.ylabel("Estimated expected value")
plt.title("Monte Carlo convergence of E[X]")
plt.legend()
plt.grid(True)
plt.show()

##2c
N = 5000000
np.random.seed(42)

X = np.random.uniform(0, np.pi, N)
Y = np.random.uniform(0, np.pi, N)

f = np.sin(2 * (np.sqrt(X**2 + Y**2) + X + Y))
K = np.pi**2

simulation_sizes = np.arange(100, N + 1, 100)
E_est = [K * np.mean(f[:n]) for n in simulation_sizes]

plt.figure(figsize=(8, 5))
plt.plot(simulation_sizes, E_est, label="Monte Carlo estimate")
plt.axhline(y=K * np.mean(f), linestyle="--", label=f"Final estimate{K* np.
    mean(f):.4f}")
plt.xlabel("Number of simulations")
plt.ylabel("Estimated integral")
plt.title("Monte Carlo convergence of double integral")
plt.legend()
plt.grid(True)
plt.show()

# task 6
N=100000
urn = ['B']*5 +['R']*5 + ['w']*5
two_ditinct = 0
for i in range(N):
    balls = np.random.choice(urn, size =3, replace=False)
```

```python
    if len(set(balls)) ==2:
        two_ditinct += 1


prob_estimate = two_ditinct / N
print (prob_estimate)
```